Peerubandi Jiten Ashrith

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Education

GITAM University 2023 – 2027
B.Tech in Computer Science Engineering (Artificial Intelligence & Machine Learning) CGPA: 8.7

Indian Institute of Technology Madras 2023 – Present
Diploma in Data Science and Programming CGPA: 6.3

Technical Skills

Languages: Python, SQL, Java, C++, C, C#
AI & Machine Learning: LLM Pipelines, RAG, LangGraph, Multi-Agent Systems, Scikit-learn, Pandas, NumPy, Model Evaluation, OpenCV, YOLOv8, SAHI, Kalman Filtering
Backend & APIs: FastAPI, REST APIs, Celery, Asynchronous Programming, PostgreSQL, MySQL
Tools & Core CS: Unity, VR Development (Meta Quest), Docker, Git, GitHub, Linux, Data Structures, Algorithms, Object-Oriented Design (OOD)

Experience

Centre for Extended Reality (CXR) Jun 2026 – Present
AI / Software Engineering Intern

- Improved target verification throughput by **25%** and reduced false positives by building **statistical validation pipelines** that measure localization accuracy and confidence scores for **TARIS**, an AI-powered multi-lane shooting range scoring system.
- Designed an **Object-Oriented multi-stage verification pipeline** with weighted confidence fusion and audit trails, processing high-throughput video streams reliably.
- Developed **FastAPI** REST services and real-time validation workflows with dashboard components for low-latency scoring and telemetry monitoring.
- Enabled quantitative debugging and **performance benchmarking** by instrumenting the computer vision pipeline to log registration, homography, and alignment metrics.

Centre for Extended Reality (CXR) May 2026 – Jun 2026
VR Developer Intern & Team Lead – VR Chemistry Laboratory

- Led a team building **VR Chemistry Laboratory**, an immersive virtual lab in **Unity (C#)** for **Meta Quest 2/3**.
- Built the entire lab environment and all interactive objects, and developed the liquid shaders used in the experiments.
- Implemented five virtual experiments: flame test, three NaOH experiments, and VIBGYOR; debugged teammates' errors.

Projects

Multi-Agent AI Critical Command Crisis Center 2026

- Architected an **LLM**-based multi-agent platform that analyzes high-volume incident data, prioritizes threats, and generates automated real-time response strategies.
- Implemented agent orchestration, **RAG** knowledge retrieval, and state management using LangGraph, Docker, and PostgreSQL.
- **Tech Stack:** Python, FastAPI, Multi-Agent AI, LLMs, RAG, LangGraph, PostgreSQL, Docker

Meeting-to-Action Agent 2026

- Built an **agentic AI** application that turns meeting content into structured, actionable follow-ups, orchestrating the workflow with **LangGraph** behind a **FastAPI** backend and a Streamlit interface.
- **Tech Stack:** Python, LangGraph, FastAPI, Streamlit, LLMs

Autonomous PTZ Aircraft Tracking System 2026

- Built a real-time perception pipeline (**YOLOv8, SAHI, Kalman filtering**) that holds stable trajectories on noisy camera feeds with sub-second response.
- Architected distributed FastAPI microservices for ONVIF camera integration, telemetry streaming, and asynchronous PTZ command scheduling using PostgreSQL and Docker.
- **Tech Stack:** Python, FastAPI, OpenCV, YOLOv8, SAHI, Kalman Filter, ONVIF, Docker, PostgreSQL

Leadership

Association for Computing Machinery (ACM) Student Chapter Jun 2026 – Present
Vice President

- Led technical workshops and coding sessions on Data Structures, Algorithms, and System Design for 200+ students, collaborating with industry speakers and academic bodies.